Journal of Neuroscience Research

Journal of Neuroscience Research 89:1897–1904 (2011)

Mini-Review

The "B Space" of Mitochondrial Phosphorylation

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It was recently shown that, in progressively depolarizing mitochondria, the F_0-F_1 ATP synthase and the adenine nucleotide translocase (ANT) may change directionality independently from each other (Chinopoulos et al. [2010] FASEB J. 24:2405). When the membrane potentials at which these two molecular entities reverse directionality, termed *reversal potential* (Erev), are plotted as a function of matrix ATP/ADP ratio, an area of the plot is bracketed by the Erev_ATPase and the Erev_ANT, which we call "B space". Both reversal potentials are dynamic, in that they depend on the fluctuating values of the participating reactants; however, Erev_ATPase is almost always more negative than Erev_ANT. Here we review the conditions that define the boundaries of the "B space". Emphasis is placed on the role of matrix substrate-level phosphorylation, because during metabolic compromise this mechanism could maintain mitochondrial membrane potential and prevent the influx of cytosolic ATP destined for hydrolysis by the reversed F_0-F_1 ATP synthase.

Key words: ANT; F_0-F_1 ATP synthase; oxidative phosphorylation; succinyl-CoA ligase; phosphate; systems biology; substrate-level phosphorylation

The mitochondrial F_0-F_1 ATP synthase is a reversible molecular machine, able to synthesize and to hydrolyze ATP (Rouslin et al., 1986, 1990; McMillin and Pauly, 1988; Petronilli et al., 1988; Wisniewski et al., 1993; Walker, 1994; Boyer, 2002). The directionality of F_0-F_1 ATP synthase is dictated by the magnitude of the protonmotive force, *pmf* (Feniouk and Yoshida, 2008).

The *pmf* consists of a transmembrane potential difference ($\Delta\Psi$m) and a transmembrane proton concentration gradient (ΔpH). Additional drive for ATP synthesis by the F_0-F_1 ATP synthase is provided by the locally high proton concentration on the cytoplasmic leaflet of the inner mitochondrial membrane (P side), possibly arising from lateral diffusion of protons ejected by the complexes of the electron transport chain along the membrane surface (von Ballmoos et al., 2009). However, in the presence of sufficiently high concentration of inorganic phosphate, as is the case in vivo (Veech et al., 1986; Katz et al., 1989; Wu et al., 2007), ΔpH is very small (<0.15; Klingenberg and Rottenberg, 1977; Vajda et al., 2009; Chinopoulos et al., 2009), so the directionality of F_0-F_1 ATP synthase is essentially controlled by $\Delta\Psi$m. The value of $\Delta\Psi$m at which F_0-F_1 ATP synthase shifts from ATP-forming to ATP-consuming is termed "reversal potential" (Erev_ATPase) and is governed by the concentrations of the participating reactants (Chinopoulos et al., 2010). At $\Delta G = 0$, by thermodynamic deduction Erev_ATPase can be calculated as follows:

$$E_{\mathrm{rev\_ATPase}} = -(316/n) - \left(\frac{2.3RT}{F}/n\right) . \log[(([\mathrm{ATP}^{4-}]free_{\mathrm{in}}.K_{M(ADP)})/((([\mathrm{ADP}^{3-}]free_{\mathrm{in}}.K_{M(ATP)} .[\mathrm{P}^{-}]_{\mathrm{in}})] - \tfrac{2.3RT}{F}.(\mathrm{pH}_{\mathrm{out}} - \mathrm{pH}_{\mathrm{in}}) \quad (3)$$

and

$$[\mathrm{P}^{-}]_{\mathrm{in}} = [\mathrm{P}_{\mathrm{total}}]_{\mathrm{in}}/(1 + 10^{\mathrm{pH_{in}} - \mathrm{pK_{a2}}}) \quad (2)$$

where in and out signify inside and outside of the mitochondrial matrix, respectively, n is the H^+/ATP coupling ratio (Watt et al., 2010), R is the universal gas constant 8.31 $J \cdot mol^{-1} \cdot K^{-1}$, F is the Faraday constant 9.64×10^4 $C \cdot mol^{-1}$, T is temperature (in degrees Kelvin), $K_{M(ADP)}$ and $K_{M(ATP)}$ are the true affinity constants of Mg^{2+} for ADP and ATP valued $10^{-3.198}$ and $10^{-4.06}$, respectively, according to Martell and Smith (1974), and $[P^-]$ is the free phosphate concentration (in molar) given by equation 2, where $pK_{a2} = 7.2$ for phosphoric acid.

ATP for the hydrolyzing F_1-ATPase of the F_0-F_1 ATP synthase complex could originate from matrix substrate-level phosphorylation (Hutson et al., 1977; Van

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Received 14 December 2010; Revised 9 February 2011; Accepted 10 March 2011

Published online 3 May 2011 in Wiley Online Library (wileyonlinelibrary.com). DOI: 10.1002/jnr.22659

et al., 1977; Chinopoulos and Adam-Vizi, 2010) or cytosolic ATP reserves (Chinopoulos and Adam-Vizi, 2010); for the latter pool to be tapped, the ANT must also reverse (Chinopoulos et al., 2010). By a minor contribution, the ATP-Mg^{2+}/P_i carrier could also provide cytosolic ATP to the hydrolyzing F_1-ATPase (Aprille, 1993). The ANT exhibits its own reversal potential (Erev_ANT), which in turn is governed by the participating reactants (Metelkin et al., 2009):

$$E_{\text{rev\_ANT}} = \frac{2.3RT}{F} \cdot \log[([ADP^{3-}]free_{\text{out}} \cdot [ATP^{4-}]free_{\text{in}}) / ([ADP^{3-}]free_{\text{in}} \cdot [ATP^{4-}]free_{\text{out}})] \quad (3)$$

The ANT utilizes the free forms of ADP^{3-} and ATP^{4-}, but the F_0-F_1 ATP synthase complex utilizes the Mg^{2+}-bound forms, $MgATP^{2-}$ and MgADP (Senior et al., 2002). However, $MgATP^{2-}$ can be expressed as $[ATP^{4-}]free \cdot [Mg^{2+}]/K_{M(ATP)}$, and MgADP can be expressed as $[ADP^{3-}]free \cdot [Mg^{2+}]/K_{M(ADP)}$. Therefore, the reversal potential of the F_0-F_1 ATP synthase and that of ANT can be expressed as functions of matrix-free ATP^{4-} and ADP^{3-}. From this, it becomes evident that alterations in $[ATP^{4-}]$ $free_{in}$ and $[ADP^{3-}]$ $free_{in}$ affect Erev_ATPase and Erev_ANT inversely; a decrease in $[ATP^{4-}]$ $free_{in}$/$[ADP^{3-}]$ $free_{in}$ ratio would shift Erev_ATPase toward more positive and Erev_ANT toward more negative values and vice versa (see below). The realization that Erev_ATPase and Erev_ANT are inversely affected by $[ATP^{4-}]$ $free_{in}$ and $[ADP^{3-}]$ $free_{in}$ could also be intuitively derived: the greater the matrix [ATP], which is the product of a forward-operating F_0-F_1 ATP synthase, the *more* it will seek to reverse; by the same token, the greater the matrix [ATP], the greater the force for a forward-operating ANT (expelling ATP from the matrix in exchange for ADP), the *less* will the ANT seek to reverse. The rest of the parameters participating in these equations may exhibit wide fluctuations; phosphate (Wu et al., 2008), the coupling ratio n (Cross and Muller, 2004; Tomashek and Brusilow, 2000), $\Delta\Psi$m ("$\Delta\Psi$m flickering"; the latter could be greater than 100 mV; Duchen et al., 1998; O'Reilly et al., 2003; Gerencser and Adam-Vizi, 2005), and of course cytosolic [ATP] and [ADP] (Yaniv et al., 2010).

COMPUTATIONAL ESTIMATIONS OF EREV_ATPASE AND EREV_ANT

The concentration of free ATP^{4-} ([L]) can be obtained by equation 4:

$$[L] = \left([L]_t \middle/ \left(1 + \frac{[Mg^{2+}]_{\text{free}}}{k_{\text{M,app}}}\right)\right) \middle/ \left(1 + \frac{10^{(-\text{pH})}}{k_{\text{H}}}\right), \quad (4)$$

where $[L]_t$ is the total ATP concentration, i.e., $[ATP^{4-}]$ + $[ATP\text{-}H^{3-}]$ + $[ATP\text{-}Mg^{2-}]$ + $[ATP\text{-}H\text{-}Mg^{-}]$; K_H is the dissociation constant for the reaction $ATP\text{-}H^{3-} \leftarrow\rightarrow ATP^{4-} + H^+$; and $K_{M,app}$ is the apparent dissociation constant of MgATP measured at pH 7.25 and 37°C (Chinopoulos et al., 2009). Similarly, the concentration of free ADP^{3-} can be obtained using equation 4 ([L], free $[ADP^{3-}]$; $[L]_t$, total ADP concentration, i.e., $[ADP^{3-}]$ + $[ADP\text{-}H^{2-}]$ + $[ADP\text{-}Mg^{-}]$ + [ADP-H-Mg]; $K_{M,app}$ is the apparent dissociation constant of MgADP measured at pH 7.25 and 37°C; Chinopoulos et al., 2009). The values for K_H and $K_{M,app}$ might be difficult to determine for the conditions found inside the matrix, and it is assumed that they are not different from those found outside the matrix. The results of the computations of equations 1, 2, 3 and 4 are depicted in Figure 1A: Erev_ATPase (black triangles) and Erev_ANT (white triangles) are plotted as functions of matrix ATP and ADP (in ratio). In this plot, the two curves define four spaces, A, B, C, and D. Mitochondria exhibiting a matrix ATP/ADP ratio for a given $\Delta\Psi$m that would place them anywhere within the A space have their ANT and F_0-F_1 ATP synthase operating in forward mode; i.e., ANT brings ADP into the matrix in exchange for ATP, and F_0-F_1 ATP synthase combines ADP and P_i to generate ATP. Mitochondria exhibiting values that would place them anywhere within the C space have both their ANT and F_0-F_1 ATP synthase operate in reverse mode; i.e., ANT brings ATP into the matrix in exchange for ADP, and F_0-F_1 ATP synthase hydrolyzes ATP to ADP and P_i. Mitochondria exhibiting values that would place them anywhere within the B space have the ANT of these organelles operating in forward mode, whereas the F_0-F_1 ATP synthase operates in reverse mode. Cytosol-mimicking conditions that would place mitochondria within D space are unlikely to exist; perhaps mitochondria can be depleted of ATP by pyrophosphate (pyrophosphate exchanges with adenine nucleotides through the ANT; Kramer, 1985) and still retain a considerable membrane potential (Asimakis and Sordahl, 1981; D'Souza and Wilson, 1982); alternatively, a low matrix [ATP] combined with high $\Delta\Psi$m may be the result of oligomycin on a forward operating F_0-F_1 ATP synthase, but in this case the ATP synthase is prevented from further operation in either forward or reverse mode. In my opinion, this part of the graph exhibits no biological representation. These computer simulations have been verified by experimental results in isolated and in situ mitochondria, showing that Erev_ATPase is almost always more negative than Erev_ANT, implying that progressively depolarizing mitochondria will first exhibit reversal of the F_0-F_1 ATP synthase, followed by reversal of the ANT (Chinopoulos et al., 2010), thereby proving the existence of the B space.

DYNAMISM OF THE BOUNDARIES OF THE B SPACE

As mentioned above, both reversal potentials are dynamic in that they depend on the fluctuating values of the participating reactants; in this section, we describe

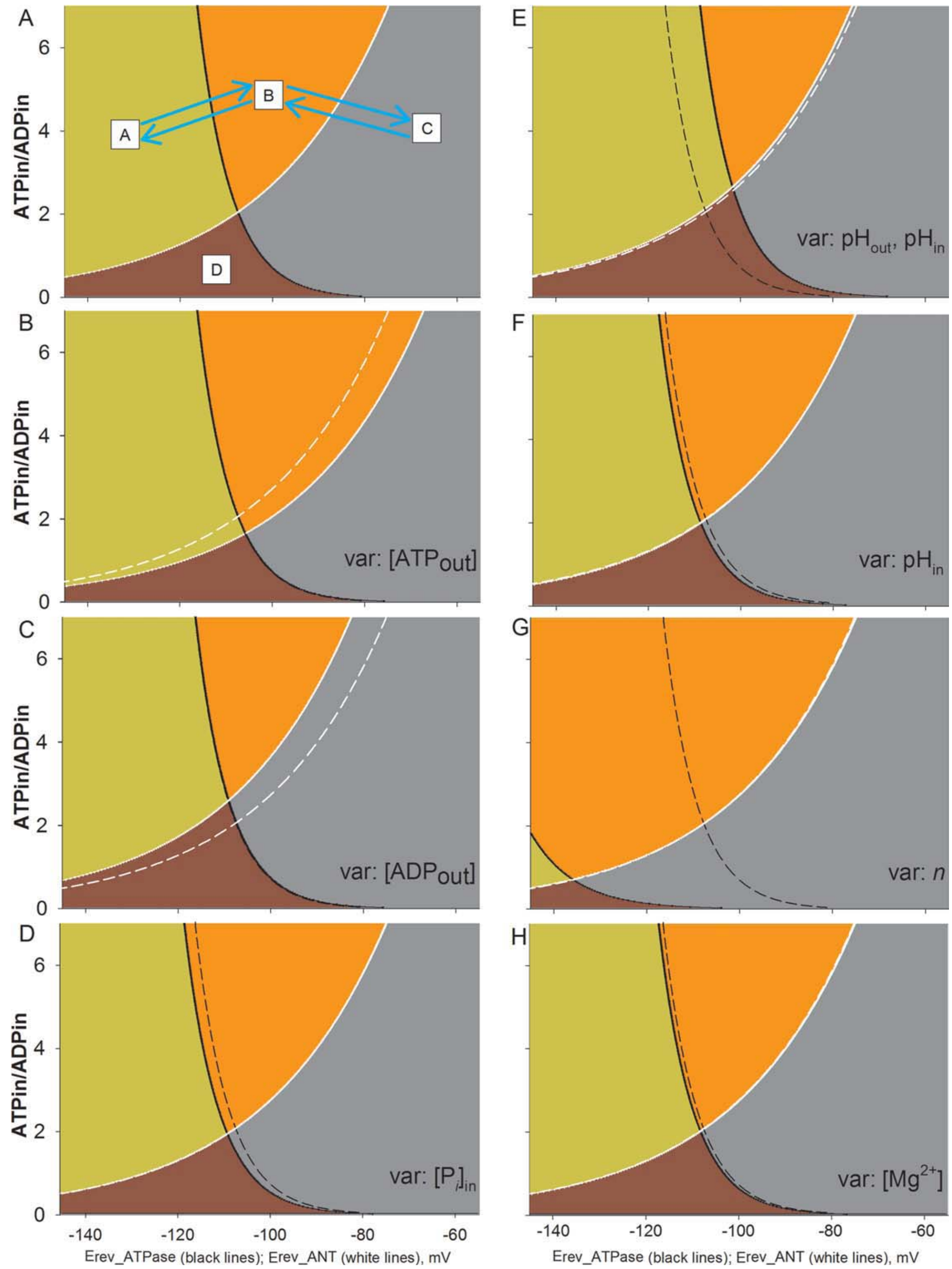

Fig. 1. **A:** Computational estimation of Erev_ANT and Erev_ATPase as a function of $[ATP]_{in}/[ADP]_{in}$ ratio. Black triangles represent the Erev_ATPase, white triangles the Erev_ANT. These values were computed for $[ATP]_{out} = 1.2$ mM, $[ADP]_{out} = 10$ μM, $P_{in} = 0.00275$ M, $n = 3.7$, $pH_{in} = 7.38$, $pH_{out} = 7.25$. Some of these values have been chosen on the basis of reports on cytosolic estimations or computational predictions (Mannella et al., 2001; Watt et al., 2010). Dashed lines for all subsequent panels represent traces of A. **B:** Variable is $[ATP]_{out}$. **C:** Variable is $[ADP]_{out}$. **D:** Variable is $[P^-]_{in}$. **E:** Variables are pH_{out} and pH_{in}, keeping ΔpH constant. **F:** Variable is pH_{in}. **G:** Variable is coupling ratio n. **H:** Variable is matrix $[Mg^{2+}]$.

how the boundaries of the B space change by altering some of the parameters found in equations 1–4. To demonstrate this, we change seven parameters individually by 25% ($[ATP]_{out}$, $[ADP]_{out}$, $[P_i]_{in}$, pH_{out}, pH_{in} (by variating ΔpH), coupling ratio n, and free matrix $[Mg^{2+}]$) unless otherwise indicated, and depict those changes in the boundaries of B space in individual panels. Dashed lines represent the positions of the boundaries prior to the 25% change. The graphs have been generated by our Windows-based software (E_{rev} Estimator), in which the user can enter any value for any parameter participating in equations 1–4, and observe the changes in Erev_ATPase and Erev_ANT. The software and instructions on how to use it can be downloaded from http://www.tinyurl.com/Erev-estimator. It cannot be overemphasized that the computational estimations elaborated below address only the directionalities of F_0-F_1 ATP synthase and ANT, not their activities.

In Figure 1B, $[ATP]_{out}$ is decreased by 25%, and this shifts Erev_ANT toward more depolarizing potentials, extending the breadth of the B space. This implies that, for in situ mitochondria undergoing progressive depolarization while the cell experiences a decrease in cytosolic [ATP], it is more likely that they will reside within the B space and be prevented from consuming cytosolic ATP (Chinopoulos et al., 2010). Erev_ATPase remains unaffected.

In Figure 1C, $[ADP]_{out}$ is decreased by 25%, and this shifts Erev_ANT toward less depolarizing potentials, thereby diminishing the breadth of the B space. This implies that, in a cell experiencing a decrease in cytosolic [ADP] harboring mitochondria that undergo progressive depolarization (e.g., by inhibition of the respiratory chain), it is less likely for the mitochondria to reside within the B space. Erev_ATPase remains unaffected.

In Figure 1D, $[P_i]_{in}$ is decreased by 25% and this shifts Erev_ATPase toward more polarizing potentials, extending the breadth of the B space but priming mitochondria to revert to ATP hydrolysis at more negative potentials. Erev_ANT remains unaffected.

In Figure 1E, both pH_{out} and pH_{in} are decreased by 25% in order to keep ΔpH constant, and this results in a migration of the B space toward more depolarizing potentials.

In Figure 1F, pH_{in} is decreased by 25%, thereby decreasing ΔpH accordingly, and this results in a migration of the B space toward more polarizing potentials; also, because the Erev_ATPase is affected by pH_{in} more than Erev_ANT, the breadth of the B space is accordingly slightly increased.

In Figure 1G, the coupling ratio n is decreased by 25%, and this shifts Erev_ATPase toward much more polarizing potentials, extending the breadth of the B space but priming mitochondria to revert to ATP hydrolysis at more negative potentials. The coupling ratio n is the same whether the F_0-F_1 ATP synthase operates in the forward or reverse mode (Turina et al., 2003; Steigmiller et al., 2008). Erev_ANT remains unaffected.

In Figure 1H, $[Mg^{2+}]$ is decreased by 25%, and this shifts Erev_ATPase toward slightly more polarizing potentials, extending the breadth of the B space but priming mitochondria to revert to ATP hydrolysis at more negative potentials. Erev_ANT remains unaffected.

A WORD OF CAUTION CONCERNING THE INTERPRETATION OF THE SHIFTS IN REVERSAL POTENTIALS RESULTING FROM ALTERATIONS IN THE PARTICIPATING PARAMETERS

From these simulations, it is apparent that alterations of some parameters would affect only one reversal potential and not the other; e.g., a decrease in $[P_i]_{in}$ would shift Erev_ATPase toward more polarizing potentials, while not affecting Erev_ANT. Intuitively, this may sound incorrect; for example, a decrease in $[P_i]_{in}$ would secondarily affect ΔpH, which in turn would affect both Erev_ATPase and Erev_ANT; furthermore, the same decrease in $[P_i]_{in}$ would affect the activity of the F_0-F_1 ATP synthase, which in turn would affect matrix [ATP] and [ADP], which in turn would affect Erev_ANT. By the same token, alterations in all parameters would ultimately affect both reversal potentials one way or another, primarily as a result of *reestablishments* of new steady states in these parameters. What is important to realize from the above simulations are two things. 1) The changes implied by the simulations address only the initial response in the alteration of one or more parameters, prior to the reestablishment of a new steady-state, and, 2) even after the reestablishments of new steady states and whatever the changes may have been, the B space still stands, albeit with altered boundaries and/or migration toward any direction of the graph (to appreciate this better, the reader is encouraged to press the button ''Automatic off'' and move any slider in the Erev_estimator software). The latter statement will hold true if the input values are realistic; e.g., the mitochondrial membrane potential cannot have a positive value, and none of the input parameters in equations 1–4 and the matrix ATP/ADP ratio can have a negative value.

WHAT IS THE FREE MATRIX ATP/ADP RATIO?

From these simulations, free matrix [ATP] and [ADP] were left out from the pool of the altered variable parameters, instead being the static variables in ratio mode. However, matrix ATP and ADP define the boundaries of the B space along the y-axis of the plot. The determination of free matrix [ATP] and [ADP] is a nearly impossible task, because of several limitations. First, if one wishes to quantify matrix ATP and ADP values during phosphorylation, this requires addition of ADP to the mitochondrial suspension, followed by conversion of ATP. That creates a technical challenge, insofar as the matrix volume is likely several thousand times smaller than the experimental volume, so matrix adenylate concentrations are many-fold lower than those appearing in the extramitochondrial compartment. Such obstacles can be addressed by centrifuging mitochondria while phosphorylating through lipid layers, thus excluding as much as possible the water-soluble extramitochondrially located nucleotides, but corrections have to be made for nucleotides residing in the intermembrane space that would be carried along the lipid layer (e.g., silicon oil). For isolated mitochondria reports of matrix ATP/ADP ratios during phosphorylation, range from 0.01 to 12 (Heldt, 1970; Heldt et al., 1972; Davis et al., 1974; Davis and Lumeng, 1974; Walajtys et al., 1974; Wieland and Portenhauser, 1974; Vignais et al., 1975; Siess and Wieland, 1976; Shrago et al., 1977; Soboll et al., 1978; Letko et al., 1980; Brawand et al., 1980; Wanders et al., 1981; Wilson et al., 1982, 1983). For mitochondria in situ or in vivo, the consensus is in the 1–3 ratio range (Soboll et al., 1980, 1984; Schwenke et al., 1981). Second, it is possible that results obtained after separation of intra- and extramitochondrial compartments are not relevant because of the time used for the separation process and possible interconversions of adenine nucleotides even in the presence of inhibitors (Pfaff and Klingenberg, 1968; Davis and Lumeng, 1974; Brawand et al., 1980; Letko et al., 1980). Third, a great part of the matrix adenine nucleotides is bound to proteins (Lusty, 1978), a notion supported by the fact that isolated mitochondria retain more than 50% of their total adenine nucleotide content after permeabilization by toluene (Matlib et al., 1977). Because of this potential binding of adenine nucleotides to intramitochondrial proteins (Harris et al., 1973; Jault and Allison, 1994; Senior et al., 2000; Boyer, 2001), the relationship between the measured total ATP/ADP ratio to *free* intramitochondrial ATP/ADP ratio is difficult if not impossible to estimate. Fourth, results from several investigators call for matrix microcompartmentation of adenine nucleotides (Vignais et al., 1975; Out et al., 1976; Vignais, 1976; Hamman and Haynes, 1983; Murthy and Pande, 1985), but this concept has not been accepted unequivocally (Heldt and Pfaff, 1969; Letko et al., 1980; Hartung et al., 1983). It is perhaps better to predict free matrix ATP/ADP ratios using mathematical modeling, depending on parameters that can be reliably quantified (Metelkin et al., 2009). By the same token, and for reasons elaborated by Metelkin et al. (2009), it is very difficult to predict the concentration of free P_i in the mitochondrial matrix. Keeping in mind the uncertainty and fluctuation of the values that dictate Erev_ANT and Erev_ATPase, it cannot be overemphasized that the actual behavior of mitochondria in terms of ATP production or consumption and in terms of whether it is cytosolic or matrix in origin is the convergent result of many complex mechanisms. Below we examine the contribution of substrate-level phosphorylation by the mitochondrial succinate-CoA ligase reaction.

CONTRIBUTION OF MATRIX SUBSTRATE-LEVEL PHOSPHORYLATION

The F_0-F_1 ATP synthase is the major player in controlling matrix ATP and ADP levels, but it is itself controlled by $\Delta\Psi$m (Feniouk and Yoshida, 2008), the variable that defines the x-axis of the graph in Figure 1. However, ATP can also be generated by substrate-level phosphorylation, a pathway that is independent from *pmf*. In the mitochondrial matrix there are two reactions capable of substrate-level phosphorylation, the mitochondrial phosphoenolpyruvate carboxykinase (PEPCK) and the succinate-CoA ligase (SUCL, or succinate thiokinase or succinyl-CoA synthetase). Mitochondrial PEPCK is thought to participate in the transfer of the phosphorylation potential from the matrix to cytosol and vice versa (Ottaway et al., 1981a; Wilson et al., 1983; Lambeth, 2002; Lambeth et al., 2004). Succinate-CoA ligase catalyses the reversible conversion of succinyl-CoA and ADP or GDP to CoASH, succinate and ATP or GTP (Johnson et al., 1998). As such, it plays a key role in the citric acid cycle (Kaufman et al., 1953), ketone metabolism (Ottaway et al., 1981b), and heme synthesis (Labbe et al., 1965) as well as being a phosphate target for the activation of mitochondrial metabolism (Phillips et al., 2009). The enzyme is a heterodimer, being composed of an invariant α subunit encoded by *SUCLG1* and a substrate-specific β subunit encoded by either *SUCLA2* or *SUCLG2*. This dimer combination results in either an ADP-forming succinate-CoA ligase (A-SUCL; EC 6.2.1.5) or a GDP-forming succinate-CoA ligase (G-SUCL; EC 6.2.1.4). The ADP-forming succinate-CoA ligase is potentially the only matrix enzyme generating ATP in the absence of a *pmf*, capable of maintaining matrix ATP levels under energy-limited conditions, such as transient hypoxia (Pisarenko et al., 1985; Weinberg et al., 2000; Holt and Riddle, 2003).

How would ADP-forming succinate-CoA ligase-mediated increase in matrix [ATP] affect the reversal potentials of F_0-F_1 ATP synthase and that of ANT? As is apparent from any panel of Figure 1, an increase in matrix ATP/ADP ratio increases Erev_ATPase and decreases Erev_ANT. Along these lines, an increase in matrix ATP/ADP ratio carries a particularly beneficial effect: assuming that a mitochondrion in a cell exhibits a $\Delta\Psi$m of -100 mV, a matrix [ATP] $=$ 4 mM, and a matrix [ADP] $=$ 2 mM, that would make an ATP/ADP ratio of 2. Assuming that all other parameters are as for Figure 1A, this mitochondrion would reside in the C space; i.e., it would consume cytosolic ATP, because both the ANT and the F_0-F_1 ATP synthase operate in reverse mode. In this case, an increase in substrate-level phosphorylation by 25% (arbitrarily chosen) will elevate matrix ATP/ADP ratio to 3.3, and that will "relocate" this mitochondrion to the B space. This mitochondrion will still consume ATP because of a reversed F_0-F_1 ATP synthase, but this ATP cannot come from the cytosolic compartment, bvecause the ANT operates in the forward mode. In this case, the F_0-F_1 ATP synthase consumes the ATP that comes from substrate-level phosphorylation, sparing the cytosolic reserves.

From the study of Chinopoulos and colleagues (2010) as well as earlier reports by the groups of Sanadi (Sanadi and Fluharty, 1963), Weinberg et al. (2000), Lambeth (Johnson et al., 1998), Chinnery (2007), and Balaban (Phillips et al., 2009), the concept is being established that matrix substrate-level phosphorylation could be a mitochondrially endogenous rescue mechanism supporting the reverse operation of F_0–F_1 ATPase, with the benefit of maintaining $\Delta\Psi$m at a suboptimal level, when mitochondria are depolarized but not sufficiently for the ANT to reverse. This assists in preserving glycolytic ATP pools that could otherwise be suppressed during prolonged ischemia (Neely and Grotyohann, 1984) but are evidently crucial for the survival chances of cells because of maintaining the function of vital ATP-dependent mechanisms, such as Na^+/K^+-ATPase and Ca^{2+}-ATPases. The importance of succinate-CoA ligase can be better appreciated by the severe phenotype observed in patients suffering from *SUCLA2* (or *SUCLG1*) deficiencies (Ostergaard, 2008). However, it would be difficult to attribute the phenotype exclusively to the provision of ATP by succinate-CoA ligase versus other very important metabolic branches, such as the maintenance of the citric acid cycle (Kaufman et al., 1953), ketone metabolism (Ottaway et al., 1981b), or heme synthesis (Labbe et al., 1965). It is also noteworthy that recent experiments by the group of Balaban (Phillips et al., 2009) showed that succinyl-CoA synthetase is activated by matrix P_i. Support for a P_i-induced activation of succinyl-CoA synthetase is also lent by the study of Siess and colleagues (1984), who determined the effects of P_i on matrix metabolite concentrations. In that study, P_i was found to increase [malate] but to decrease [α-ketoglutarate] and [succinyl-CoA], implying activation of succinyl-CoA synthetase. Along these lines, during cardiac ischemia, succinate is known to accumulate (Pisarenko et al., 1985; Pisarenko, 1996), a phenomenon that is consistent with an increase in succinyl-CoA synthetase, possibly as a result of activation by P_i. Activation of succinyl-CoA synthetase toward the formation of ATP and succinate would obviously have the implication of providing ATP for the consuming F_0–F_1 ATPase, thereby maintaining a moderate $\Delta\Psi$m that, if more negative than Erev_ANT, could contribute to sparing cytosolic ATP pools. A beneficial byproduct is the provision of matrix P_i from matrix ATP hydrolysis that could activate succinyl-CoA synthetase toward extra ATP formation. Finally, as elaborated above (Fig. 1D), an additional benefit of the elevation of [P_i] would be the priming of mitochondria to reverse ATP hydrolysis at more positive potentials, thereby rendering them less prone to matrix ATP consumption.

Having discussed the potential benefits of matrix substrate-level phosphorylation generating ATP, the question arises of whether, and to what extent, this mechanism can remain operational during metabolic compromise, given that it utilizes succinyl-CoA, which is a product of α-ketoglutarate dehydrogenase complex (KGDHC) that in turn requires provision of NAD^+. In a metabolic standstill, the respiratory complexes are inactive and do not oxidize NADH. Still, in the experiments with isolated and in situ mitochondria by Chinopoulos et al. (2010), the beneficial contribution of substrate-level phosphorylation was readily evident even after tens of minutes of complete respiratory chain inhibition. Mitochondria probed from transgenic mice exhibiting partial reductions in KGDHC activity showed that, during complete respiratory chain inhibition, provision of substrates favoring the succinyl-CoA synthetase reaction toward ATP formation is critical for shifting Erev_ANT toward more polarized $\Delta\Psi$m values (unpublished observations). That can only mean that, during respiratory chain inhibition, KGDHC remained operational, which in turn implies that NAD^+ was available. Ongoing efforts in our laboratory are focused on identifying the source of NAD^+ in the absence of oxidative phosphorylation, and at least two concepts are being explored: 1) the malate dehydrogenase reaction that can yield NAD^+ and malate by oxidizing oxaloacetate and NADH and 2) ROS-mediated NADH oxidation by KGDHC (Starkov et al., 2004; Tretter and Adam-Vizi, 2004).

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